

WESLEY LAWSON

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EDUCATION

Boston College

Bachelor of Arts in Computer Science, cum laude

Chestnut Hill, MA

Aug. 2022 – May 2026

- GPA: 3.71 / 4.00 | Major GPA: 3.83 / 4.00
- Relevant Coursework: Computer Systems, Operating Systems, Computer Networks, Software Engineering, Algorithms

EXPERIENCE

Undergraduate Teaching Assistant, Computer Science 2

Boston College Department of Computer Science

Aug. 2025 – May 2026

Chestnut Hill, MA

- Held a weekly discussion section of 15 students, leading C++ examples and live code walkthroughs
- Guided students through debugging in weekly office hours and graded 20 submissions per assignment

Software Development Intern

XYZ Homework

June 2025 – Aug. 2025

Remote

- Extended the assessment library for high-level math courses using IMathAS, PHP, and the GeoGebra API

PROJECTS

Lembas Links | Go, Gin, Redis, Postgres, Docker, AWS

Aug. 2026

- Developed a Lord of the Rings themed URL shortener as a REST API in Go using Gin and Redis
- Strengthened the API with anonymous session keys stored as SHA-256 hashes and Redis rate limiting
- Recorded click analytics asynchronously and containerized the full stack with Docker Compose
- Deployed to dual EC2 instances behind Nginx, isolating each component with security groups

BitTorrent Client | Go, TCP, Concurrency

March 2026

- Engineered a BitTorrent client in Go from scratch, parsing bencoded .torrent metainfo
- Handled tracker HTTP negotiation and TCP peer handshaking with info hash verification
- Sustained parallel downloads from multiple peers with goroutines, channels, and a worker pool
- Downloaded a full 700 MB Debian ISO from a live swarm, verifying each piece against SHA-1

Recursive DNS Server | Go, DNS, Computer Networks

Nov. 2025

- Built a recursive DNS server in Go, parsing messages at the binary level with encoding/binary
- Eliminated transaction ID collisions between concurrent clients with a mutex-guarded query map
- Answered from a parsed zone file, cached results with a TTL, and wrote Go unit tests

Xv6 Operating System Extensions | C, Kernel, Operating Systems

Jan. 2026

- Implemented xv6 kernel system calls in C, including sysinfo, ps, and process/kernel backtrace
- Redesigned the process scheduler from round-robin to a Multi-Level Feedback Queue
- Managed process priority queues and preemption logic at the kernel level

LEADERSHIP AND ACTIVITIES

Operations Coordinator, Trips and Excursions

Boston College Campus Activities Board

Sep. 2022 – May 2026

Chestnut Hill, MA

- Produced off-campus trips and campus-wide events over four years by working with vendors
- Led internal bonding experiences for the department and staffed events on site

Freshman League Captain

Boston College

Sep. 2025 – May 2026

Chestnut Hill, MA

- Guided a group of freshmen through their first year with weekly discussions on college life

TECHNICAL SKILLS

Languages: Go, C, C++, Python, SQL (Postgres), JavaScript, TypeScript, Java

Frameworks: Gin, Django, Django REST Framework, React

Developer Tools and Libraries: Linux, Git, AWS (EC2, VPC), Docker, Docker Compose, Redis, Wireshark, JWT